

# Multiphase flow predictions from carbonate pore space images using extracted network models

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[1] A methodology to extract networks from pore space images is used to make predictions of multiphase transport properties for subsurface carbonate samples. The extraction of the network model is based on the computation of the location and sizes of pores and throats to create a topological representation of the void space of three-dimensional (3-D) rock images, using the concept of maximal balls. In this work, we follow a multistaged workflow. We start with a 2-D thin-section image; convert it statistically into a 3-D representation of the pore space; extract a network model from this image; and finally, simulate primary drainage, waterflooding, and secondary drainage flow processes using a pore-scale simulator. We test this workflow for a reservoir carbonate rock. The network-predicted absolute permeability is similar to the core plug measured value and the value computed on the 3-D void space image using the lattice Boltzmann method. The predicted capillary pressure during primary drainage agrees well with a mercury-air experiment on a core sample, indicating that we have an adequate representation of the rock's pore structure. We adjust the contact angles in the network to match the measured waterflood and secondary drainage capillary pressures. We infer a significant degree of contact angle hysteresis. We then predict relative permeabilities for primary drainage, waterflooding, and secondary drainage that agree well with laboratory measured values. This approach can be used to predict multiphase transport properties when wettability and pore structure vary in a reservoir, where experimental data is scant or missing. There are shortfalls to this approach, however. We compare results from three networks, one of which was derived from a section of the rock containing vugs. Our method fails to predict properties reliably when an unrepresentative image is processed to construct the 3-D network model. This occurs when the image volume is not sufficient to represent the geological variations observed in a core plug sample.

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## 1. Introduction

[2] In the Middle East, more than 60% of the hydrocarbon reserves are contained in carbonate reservoirs [Konyuhov and Maleki, 2006]. The design of improved oil recovery in these fields requires better reservoir characterization. At the fundamental scale, pore scale, carbonate rocks can be highly irregular because of their complex geological history, including diagenesis by animal organisms and chemical reaction. An understanding of flow and transport processes in carbonates remains a challenge.

[3] Pore network models provide a tool to understand and predict single and multiphase flow and transport in porous media [Bryant and Blunt, 1992; Blunt et al., 2002; Valvatne and Blunt, 2004; Piri and Blunt, 2005a, 2005b; Øren et al., 1998; Al-Futaisi and Patzek, 2003]. To build predictive models, topologically representative pore throat networks

need to be used. The starting point for generating these networks is a three-dimensional (3-D) image of the pore space.

### 1.1. Three-Dimensional Rock Images for Carbonates

[4] Micro-CT scanning can image the pore space of rocks with a resolution of around a few microns [Dunsmuir et al., 1991]. However, most carbonate pore systems, with submicron features, lie below the resolution of CT scanning tools. It is possible to image the pore space in two dimensions using thin sections and electron microscopy. From this a statistically representative 3-D image may be generated. In this paper we will use multiple-point statistics that attempt to reproduce typical patterns in the pore space and the long-range connectivity of the rock [Okabe and Blunt, 2004, 2005; Al-Kharusi and Blunt, 2007].

### 1.2. Network Extraction

[5] Various techniques have been proposed to construct topologically representative networks representing sandstones from 3-D images [Lindquist and Lee, 1996; Lindquist and Venkatarangan, 1999; Lindquist, 2002; Vogel and Roth, 2001; Silin et al., 2003; Silin and Patzek, 2006; Arns et al., 2003, 2004a, 2004b; Al-Raoush et al., 2003; Al-Raoush and

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Willson, 2005a, 2005b; Delerue and Perrier, 2002; Delerue et al., 1999]. To date, however, the only networks that have then been used for successful predictions of multiphase flow properties are those generated from representations of the pore space derived from simulating the packing and cementation of grains, the process-based method [Øren et al., 1998; Øren and Bakke, 2003]. The networks generated by Øren and Bakke used grain size distribution information and other petrographical data obtained from 2-D sandstone thin sections. Then, knowing the grain centers, pores and throats in an equivalent network model could be identified. However, for carbonates, with a complex diagenetic history, it is not easily possible to generate a representative grain packing.

[6] Sheppard et al. [2005] used a medial axis transform pioneered by Lindquist and coworkers [Lindquist and Lee, 1996; Lindquist and Venkatarangan, 1999; Lindquist, 2002] with morphological measures to extract networks from large micro-CT images. Arns et al. [2005] imaged a range of carbonate core and outcrop material using their in-house micro-CT imaging system. The porosity and permeability computed on these 3-D images compared well with experimental data. They confirmed the large difference between the topological properties of carbonates and sandstones.

[7] In this work, we use the maximal ball concept of Silin et al. [2003] and Silin and Patzek [2006] to extract a network. This approach identifies pores by growing spheres in the void space; chains of spheres of decreasing size represent throats. We have shown that networks for both Berea and Fontainebleau sandstones have similar topological properties to those extracted from the same images using a process-based approach by Øren and Bakke [Al-Kharusi and Blunt, 2007]. We further constructed network models from carbonate subsurface samples and compared absolute permeability values with laboratory measurement and lattice Boltzmann calculations. In this paper we will construct additional carbonate networks for which we will predict multiphase flow properties and compare to experimental measurements.

### 1.3. Multiphase Fluid Flow Predictions

[8] Lattice Boltzmann techniques can be used to compute multiphase flow properties, but because of the high computational expense to simulate quasi-static displacement, their use has been limited [Gunstensen and Rothman, 1993; Chen and Doolen, 1998; Hazlett et al., 1998]. Semiempirical methods for predicting multiphase flow properties have been suggested, such as combining the *Mualem* [1976] relative permeability model with the *van Genuchten* [1980] effective capillary pressure model [Dullien, 1992]. The capillary pressure is first matched and then relative permeability is predicted. However, the method assumes a strongly water-wet medium and uses parameters that may not properly encapsulate the displacement processes and pore space geometry.

[9] Pore network models provide an elegant method to compute two and three-phase flow properties; semianalytic expressions for quasi-static flow and displacement in individual pores and throats are combined to make predictions for small samples, 10–100 pores across (a few mm) [Øren et al., 1998; Bakke and Øren, 1997; Øren and Bakke, 2003; Blunt et al., 2002; Valvatne and Blunt, 2004; Al-Futaisi and Patzek, 2003; Lerdahl et al., 2000; Piri and Blunt, 2005b;

Svirsky et al., 2007; Suicmez et al., 2007]. These models can simulate any sequence of oil, water and gas injection in elements (pores or throats) of arbitrary wettability (contact angle). Each element can have an angular cross section. This allows wetting fluids to reside, connected, in the corners while the nonwetting phase occupies the centers.

[10] One key issue is the assignment of contact angles in the model. While the network captures the geometric structure of the rock, it does not provide information about its wettability that has a significant impact on relative permeability and capillary pressure [Valvatne and Blunt, 2004]. Most, if not all, reservoir rocks are not strongly water wet. Current models account for wettability alteration due to contact of crude oil with the rock surface after primary drainage and allow the contact angles to be different for waterflooding and oil reinvasion [Øren and Bakke, 2003; Valvatne and Blunt, 2004]. In carbonates this is of particular significance since most reservoir carbonates display oil-wet characteristics [Dullien, 1992].

[11] Valvatne and Blunt [2004] predicted relative permeability for a carbonate sample using a quasi-static network model. They used a network based on Berea sandstone and modified the pore and throat size distributions to match the primary drainage capillary pressure. They then assigned contact angles to match the measured residual oil saturation. In this paper instead we use a network derived directly from a carbonate sample and so we can predict the primary drainage capillary pressure without tuning the network properties. However, we do still require a method to assign contact angles.

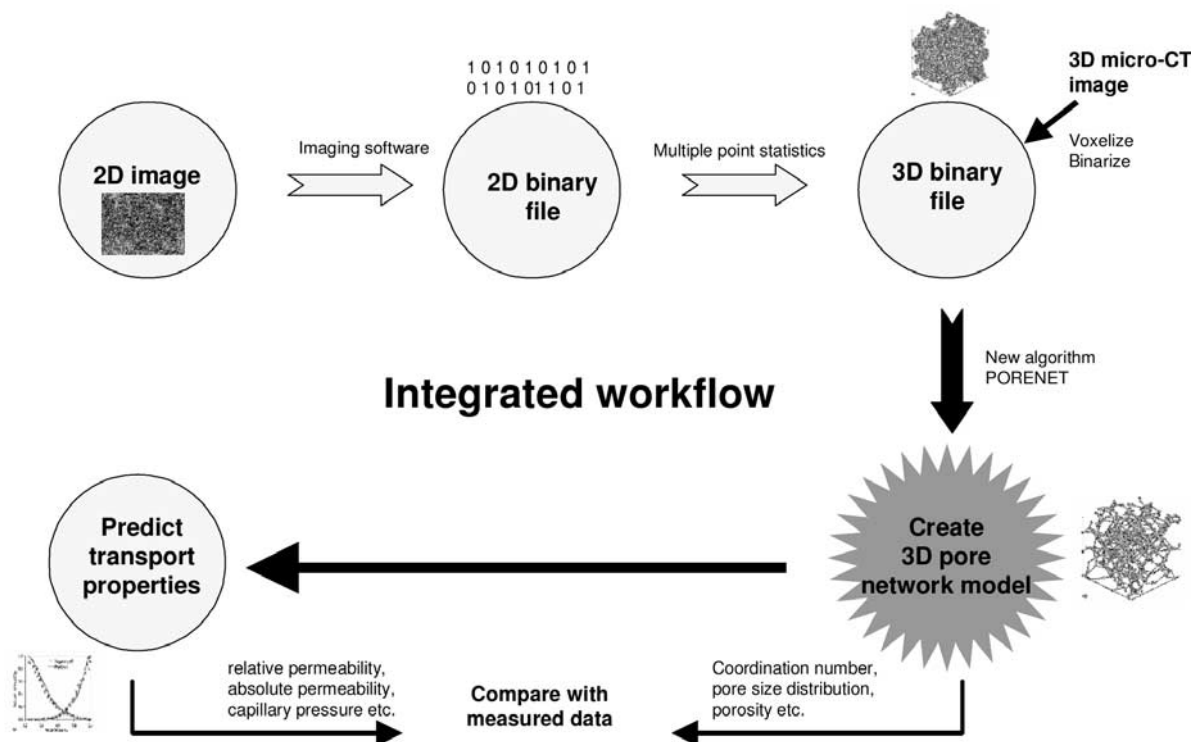
## 2. Carbonate Network Extraction and Comparison With Laboratory Data

### 2.1. Workflow for Carbonates

[12] The network extraction algorithm uses the maximal ball concept originally described by Silin et al. [2003]. A complete description of the algorithm is provided in Al-Kharusi and Blunt [2007] but the essential features are repeated here for completeness.

[13] Maximal balls are first computed: they are the largest spheres centered on any voxel in the image that just fit in the pore space. The largest maximal balls, that have no overlapping neighbors that are larger, identify pores. Throats are found by following chains of balls between these pores. Pores can be constituted from either a single maximal ball or a group of balls adjacent to or overlapping each other. Throats are made of balls decreasing in diameter, connecting two maximal balls, or a group of neighboring balls equal in size [Al-Kharusi and Blunt, 2007].

[14] The pore body and pore throat volumes are computed by referring each pore or throat to the original image voxels comprising the spheres involved. Once the volumes are determined then an equivalent effective pore body radius is calculated by equating the pore body volume obtained to the volume of an equivalent sphere. In the case of a throat, an equivalent effective throat radius is calculated by equating the throat volume obtained to an equivalent cylindrical volume. The length of the cylinder is the maximum distance computed between any two spheres comprising the throat entity. Note that these shapes are only used to find effective



**Figure 1.** The integrated multistaged workflow approach used in this study.

radii but the actual shapes assumed for flow computations are governed by the shape factors described below.

[15] Shape factors, the ratio of cross-sectional area of an element to perimeter squared, are assumed to have a truncated Weibull distribution ranging from a sharply edged triangular shape (having a shape factor value of 0.01) to an equilateral triangular shape (shape factor of 0.048) with a standard deviation of 0.008. In this study, investigation of shape factors of half that of an equilateral triangle (0.024) has also been looked at, refer to Section 2.5. A proportion of the pore space to represent clays that always remain full of water, is assigned to the model; it is given a fixed, small value of 0.5% to match the measured irreducible water saturation in the experiments.

[16] The extracted network is input into a two-phase pore-scale simulator, originally developed by *Valvatne and Blunt* [2004]. Absolute permeability and formation factors are calculated when the network is fully saturated with a single phase. For comparison purposes, absolute permeability was also directly computed on the 3-D image using the lattice Boltzmann method [*Al-Kharusi and Blunt*, 2007; *Okabe and Blunt*, 2004].

[17] In this study, a multistaged workflow was implemented where we started from easy-to-obtain data (a 2-D image) and finished by predicting difficult-to-measure data such as relative permeability. This is achieved in five different stages (Figure 1). The first stage is to acquire a representative 2-D image of the void space. This is done by imaging a thin section of the rock using electron microscopy at a resolution of less than  $1\ \mu\text{m}$ . The second stage is to threshold and then to digitize the image. The third stage is to convert the image into a 3-D representation of the pore space. The fourth stage is to extract a topologically representative network model of pores and throats from the 3-D

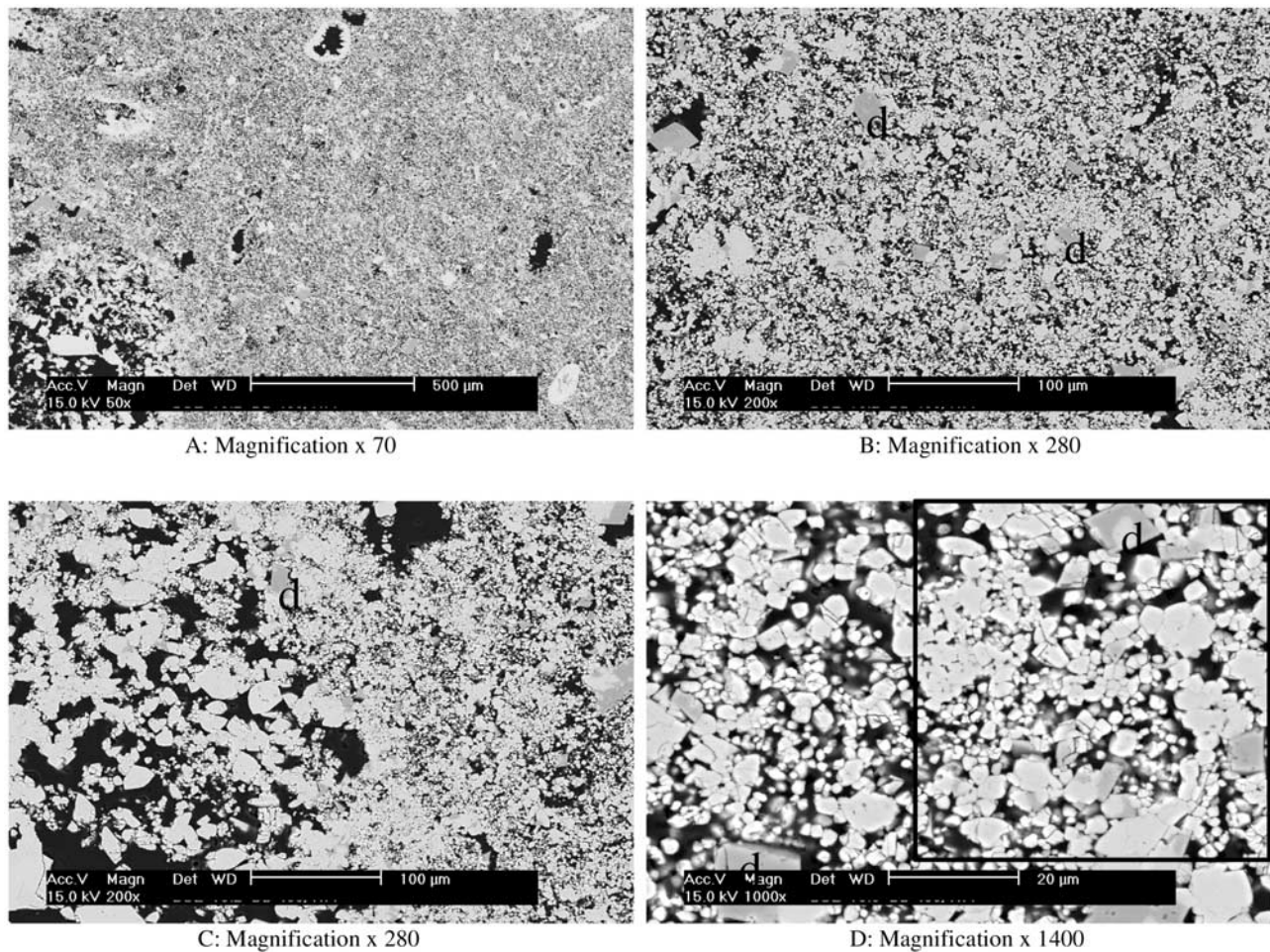
image. Finally we input the extracted network into a pore-scale flow simulator to compute multiphase transport properties. At the end of this section we compare predictions from three networks extracted from different core plugs.

## 2.2. Construction of the Pore Network Model

[18] A cretaceous subsurface carbonate core plug from a giant Middle Eastern field containing light oil (viscosity  $1.4\ \text{mPa}\cdot\text{s}$ ) was chosen for the analysis. A thin section was cut and imaged at various resolutions using scanning electron microscopy, SEM, (Figure 2) for the core plug investigated. Two-dimensional high-resolution SEM images are used here as we cannot obtain 3-D images with the required resolution to image all the pore space for this carbonate sample. The plug chosen contains few vugs and has a relatively homogeneous pore structure.

[19] Figure 2 illustrates the characteristics of the pore system in this compacted and leached peloidal packstone (Figure 2a). Note the presence of some isolated vugs/molds (in solid black) scattered in a predominantly microporous matrix (microporosity is shown in gray). Granular components are not distinguishable any more (Figure 2b), probably because of the combined effect of compaction, pervasive leaching and recrystallization. Dolomite crystals (marked d) are scattered throughout the matrix, postdating calcite. Figure 2c shows a detail taken at the boundary between a dissolution vug and the matrix. This shows that the mesopore within the vug is surrounded by, and connected to, a porous system dominated by micro pores. These are generally well connected at the microscopic scale (Figure 2d).

[20] An image with sufficient resolution to observe the majority of the pore space was selected (Figure 3). The image was thresholded into black and white and then

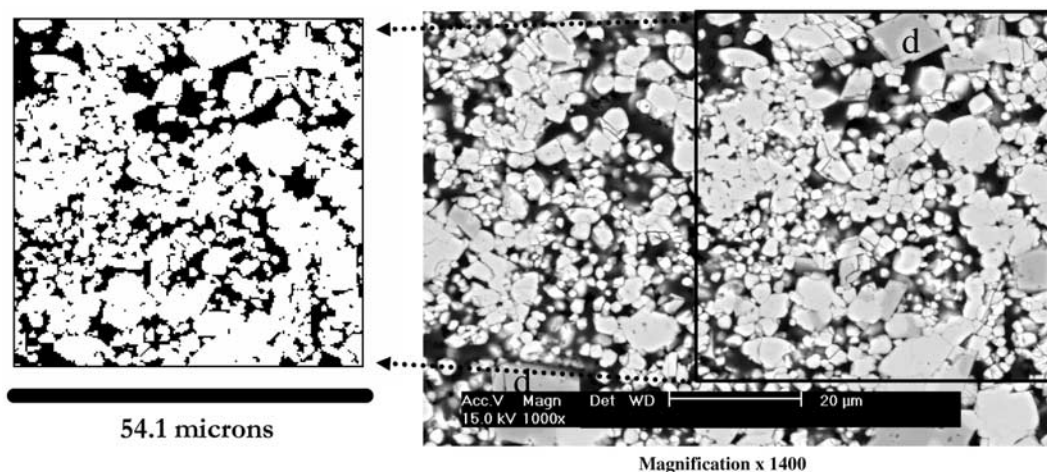


**Figure 2.** (a–d) SEM images at different resolutions taken from a carbonate subsurface core plug. The square in Figure 2d highlights the region used to generate a 3-D image of the pore space. This sample was chosen as it has few vugs and a relatively homogeneous pore structure.

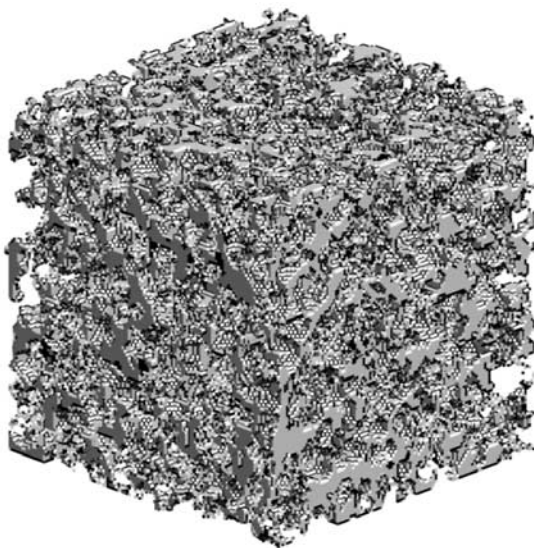
digitized. The image used in this analysis is of size  $54.1 \mu\text{m}$ , taken at a resolution of  $0.27 \mu\text{m}$  ( $200^2$  total voxels). The image at this stage is still two dimensional. Note that for this carbonate, the pore space could not readily be resolved

using micro-CT scanning and therefore no 3-D image is available.

[21] The digitized 2-D image above was input into an algorithm developed by *Okabe and Blunt* [2004] that uses



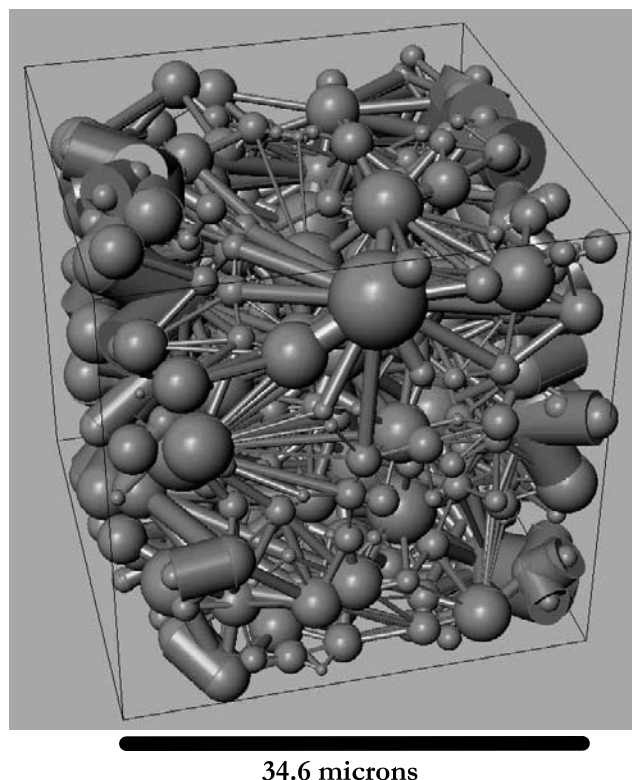
**Figure 3.** Two-dimensional SEM image of the carbonate sample, from which the 3-D network was extracted (black areas denote pore spaces; image was digitized into 200 by 200 voxels with a resolution of  $0.27 \mu\text{m}$ ).



**Figure 4.** Three-dimensional carbonate image obtained using multiple-point statistics ( $128^3$  voxels).

multiple-point statistics to construct a digitized 3-D image through reproducing the same patterns seen in the 2-D image. It is assumed that the image is isotropic. The resultant 3-D image is given in Figure 4.

[22] The 3-D image was then input into the network extraction code, generating a topologically representative network of the carbonate rock. The carbonate network contains 643 pores and 2623 throats. The network is shown



**Figure 5.** The extracted pore network for the carbonate sample obtained from the image shown in Figure 4. The network contains 643 pores and 2623 throats.

**Table 1.** Static Properties of the Carbonate Network Model

| Property                        | Value                 |
|---------------------------------|-----------------------|
| Image size ( $\mu\text{m}$ )    | 34.6                  |
| Image volume ( $\text{mm}^3$ )  | $4.14 \times 10^{-5}$ |
| Image total voxels              | $128^3$               |
| Number of pores                 | 643                   |
| Number of throats               | 2623                  |
| Average connection number       | 7.9                   |
| Number of connection at inlet   | 41                    |
| Number of connections at outlet | 64                    |
| Net porosity (%)                | 29                    |
| Absolute permeability (mD)      | 3.1                   |
| Formation factor                | 7.2                   |

in Figure 5. The network extraction ran in approximately 5 days and reached a maximum processing memory of approximately 16 GB on a Linux cluster. The multiphase flow simulations in contrast took less than a minute to run.

### 2.3. Comparison of Predicted and Measured Static Properties

[23] The network properties are given in Table 1. Table 2 gives a comparison of the absolute permeability predicted by the carbonate network, computed using the lattice Boltzmann method and the experimentally measured value using the core plug that the image comes from: there is a reasonably good agreement between the network-predicted permeability and that measured on a larger core plug.

[24] In Figure 6, the pore and throat radii are plotted against the number of network elements. As expected the throats have much smaller radii than the pores; the pores are well connected and, on average, are connected to eight other pores. In contrast, sandstones, where the pores and throats are of comparable size also typically have a lower coordination number, of around 4 or lower [Øren and Bakke, 2003]. Also note that all the throats have radii less than  $1 \mu\text{m}$ ; these elements could not be imaged directly in 3-D using micro-CT scanning at current resolutions.

### 2.4. Comparison of Predicted and Measured Multiphase Transport Properties

[25] Below we will briefly summarize the experiments conducted to measure the various two-phase flow parameters for this field. We will then compare the network model results with the experimental results for the same core plug.

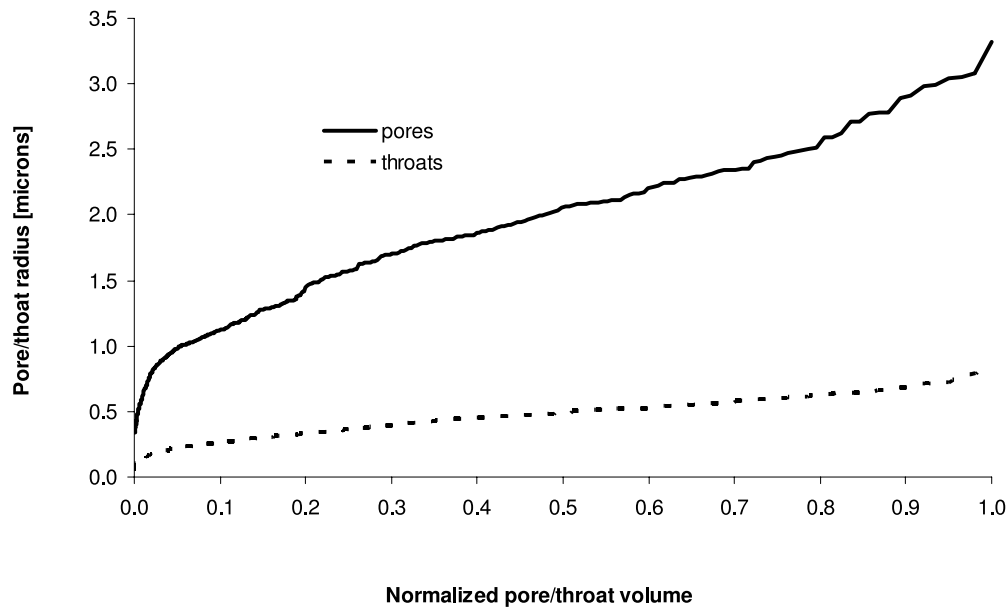
#### 2.4.1. Laboratory Experiments Conducted

[26] For this reservoir, a comprehensive special core analysis (SCAL) program was conducted and many samples were analyzed to capture the large variation in this hetero-

**Table 2.** Comparison of Predicted and Computed Versus Measured Absolute Permeability<sup>a</sup>

|                                        | Predicted<br>Absolute<br>Permeability | Computed<br>Absolute<br>Permeability | Experimental<br>Absolute<br>Permeability |
|----------------------------------------|---------------------------------------|--------------------------------------|------------------------------------------|
| Absolute permeability<br>to brine (mD) | 3.1                                   | 1.9                                  | 3.4                                      |

<sup>a</sup>Predicted permeability is from the network model (this work), computed permeability is from the lattice Boltzmann method, and experimental permeability is from the core plug experiment.

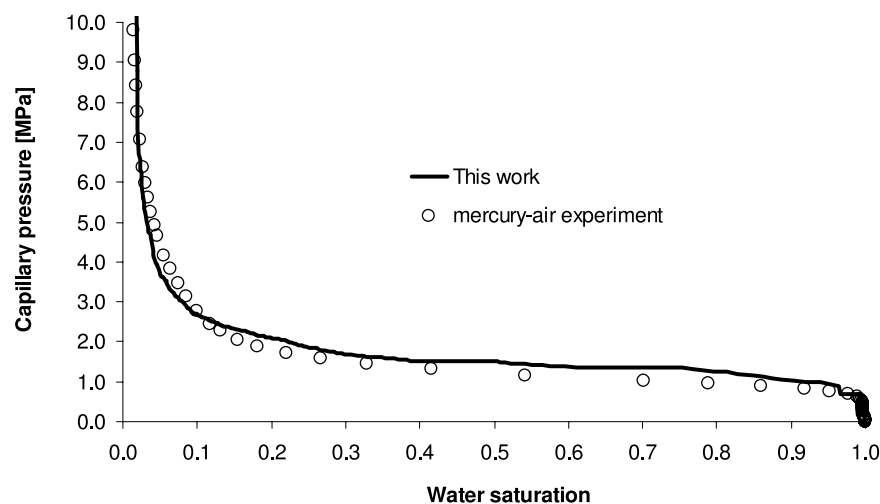


**Figure 6.** Pore and throat radii plotted against network element number.

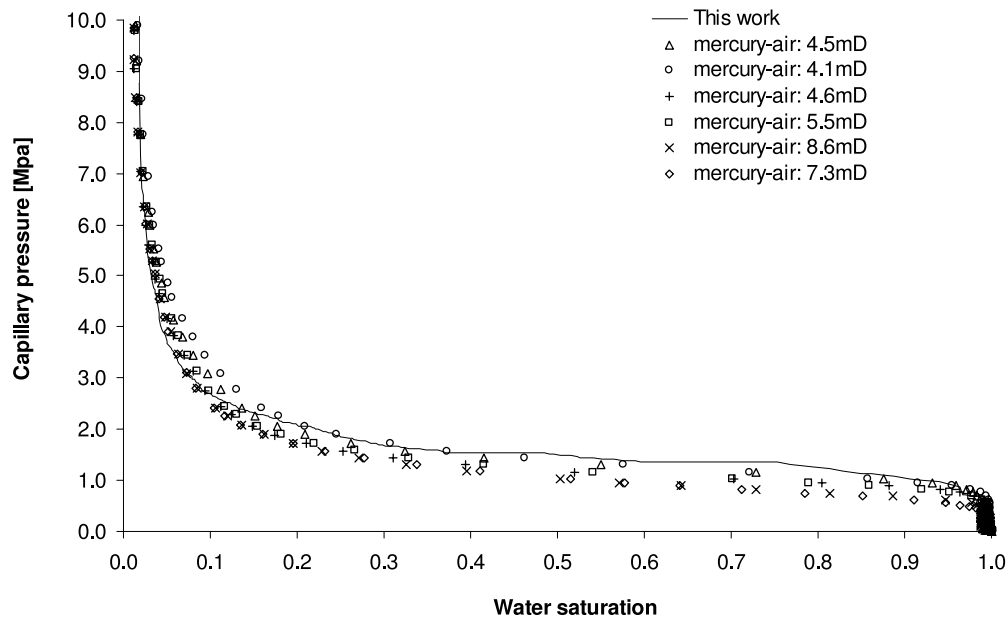
geneous carbonate field. The permeability varies over four orders of magnitude ranging from less than a milliDarcy to more than a Darcy. The porosity of the field ranges between 10 to over 30%; however, most of the initial oil resides in rock types with porosity in a range between 20 and 30%. Core plugs of size 3.78 cm in width and 4.90 cm in length were drilled for various experiments. 250 samples were taken for porosity and permeability measurement. A subset of 80 samples was selected from different permeability and porosity ranges representing key geological facies for the subsequent SCAL program, the details of which have been described by *Masalmeh and Jing* [2004]. The program includes rock characterization, CT scanning, NMR measurements, mercury-air capillary pressure, and analysis of scanning electron microscope thin-section images. The program also includes oil/water primary drainage capillary pressure

measurements, aging the samples in crude oil at irreducible water at reservoir temperatures (120°C) and pressures (27 Mpa) for four weeks to restore wettability and then conducting spontaneous imbibition measurements followed by waterflood and secondary oil injection centrifuge experiments.

[27] 50 samples were selected for mercury-air and water-oil centrifuge capillary pressure measurements. We selected a plug that has a representative value of porosity and permeability for one permeability class: 1–10 mD. The investigation of other permeability classes will be the subject of further work. For mercury injection, mercury is always the nonwetting phase, whereas for the centrifuge experiments, the wettability of the sample represents that of the oil and brine at field conditions. The average saturation in a core plug is recorded at a set of centrifugal speeds and



**Figure 7.** Predictions using the carbonate network compared with laboratory measurements of primary drainage capillary pressure (mercury-air system).



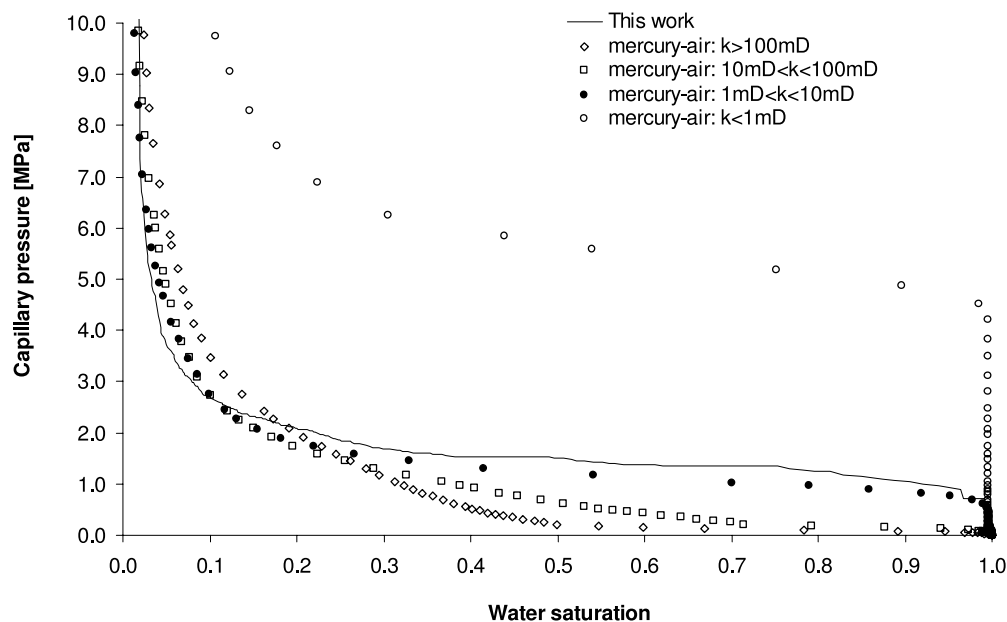
**Figure 8.** Network-predicted versus measured primary drainage capillary pressure: six core plugs with a similar permeability range (mercury-air system).

capillary pressure versus saturation point is calculated at each speed. The capillary pressure can be calculated from the centrifuge raw data using either analytical [Forbes, 1997] or numerical methods [Maas and Schulte, 1997]. Numerical simulations using MoReS, the Shell in-house simulator, were used to model the experimental results and the capillary pressure and displaced-phase relative permeability were found that best fit the measurements [Masalmeh and Jing, 2004]; the centrifuge measurements do not measure the relative permeability of the injected phase.

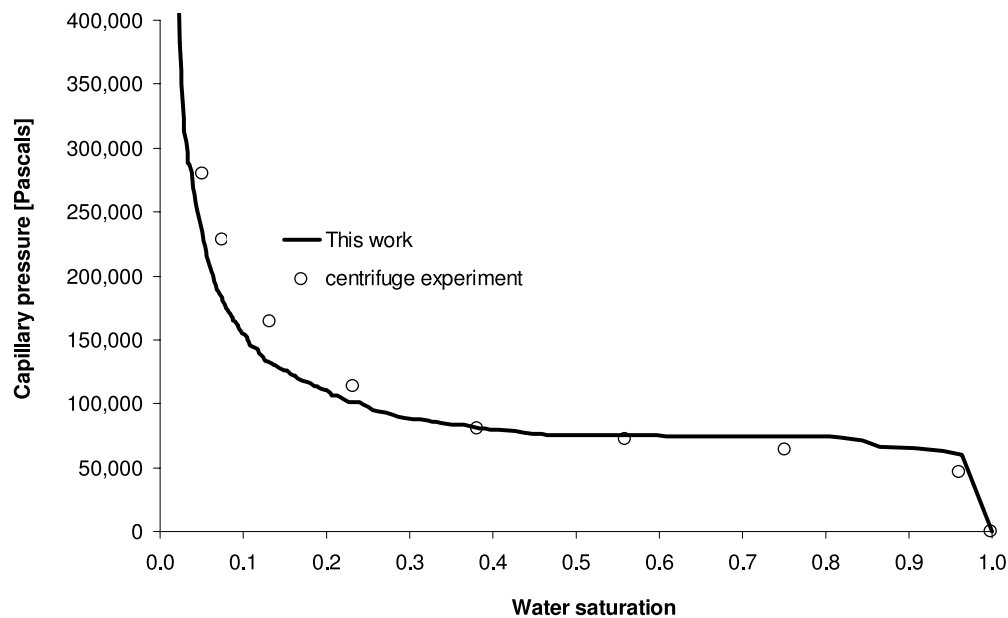
[28] Waterflood capillary pressure was measured for 32 samples. No spontaneous imbibition of water was observed on any of the samples following aging with crude. This indicates that all the carbonate samples had oil-wet characteristics.

#### 2.4.2. Two-Phase Fluid Flow Predictions

[29] The carbonate network generated above is now input into a two-phase, public domain pore-scale simulator [Valvatne and Blunt, 2004] to predict multiphase fluid flow properties. We first predict mercury-air primary drainage



**Figure 9.** Network-predicted versus measured primary drainage capillary pressure: four core plugs from different permeability classes (mercury-air system).



**Figure 10.** Comparison of network-predicted versus measured primary drainage capillary pressure (brine-oil centrifuge experiment).

and compare with the experimental data for the chosen core plug. The core plug we analyzed has a porosity of 29% and an absolute permeability to brine of 3mD. We input a mercury-air system in our carbonate network model with mercury-air interfacial tension of 485 mN/m and a receding contact angle of  $40^\circ$  [Valvatne and Blunt, 2004]. We plot the comparison in Figure 7.

[30] In Figure 8 we show mercury-air primary drainage capillary pressure curves for core plugs of similar permeability from the same field. In Figure 9 we show the curves for core plugs of different permeability classes highlighting the heterogeneity in this field. Note that the capillary pressures for different permeability classes are quite distinct and that our predictions are in the range of behavior seen for the selected permeability with only a slight overestimation of capillary pressure at low saturation.

[31] The fluid properties are then modified to represent the conditions of the displacement experiments. A reservoir brine-oil interfacial tension of 22.5 mN/m, receding contact angle of  $30^\circ$  in primary drainage, brine viscosity of 0.66 mPa s, oil viscosity of 1.4 mPa s, brine density of 1098 kg/m<sup>3</sup> and oil density of 782 kg/m<sup>3</sup> are used. The contact angle was chosen on the basis of the core analysis of Masalmeh and Jing [2004]. We now calculate primary drainage capillary pressure and relative permeability using the brine-oil system.

[32] Relative permeability was only measured on a few core samples. In this section we will compare against one set of measurements for the core plug whose capillary pressure is shown in Figure 7. The capillary pressure and relative permeability predictions compared with experiment for primary drainage are given in Figures 10 and 11.

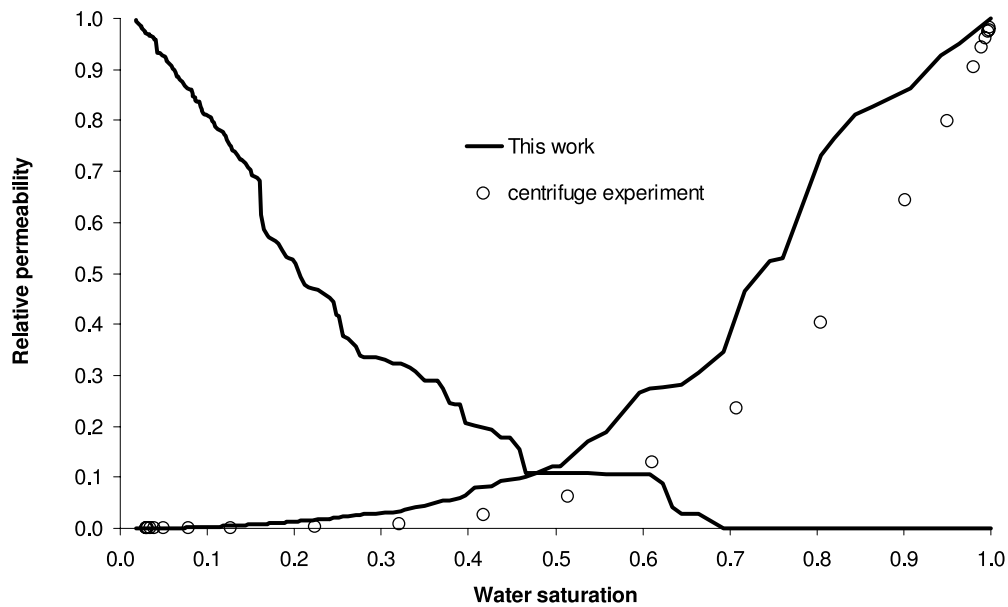
[33] As expected, the network-predicted capillary pressure using a brine-oil system compares well with the centrifuge experiment. Physically, this is an identical displacement process to the mercury-air capillary pressure shown in Figure 7, except for a scaling to account for

interfacial tension and contact angle. Our extracted network captures the pore size distribution and makes good predictions in this case.

[34] The predicted water relative permeability is overestimated compared to the experimental results, although the trend and shape of the curve is correct; the oil relative permeability was not measured. We may overestimate the connectivity of the medium in our network extraction algorithm giving a large coordination number, leading to an overprediction of relative permeability [Al-Kharusi and Blunt, 2007]. Furthermore, we use a very small network that may not be representative of the whole core sample. The good prediction of capillary pressure implies that we have estimated pore and throat sizes correctly, but we would appear to assign too high a flow rate to those larger elements occupied by the nonwetting phase.

[35] Following primary drainage, we inject water in the network. For waterflooding, we modify the contact angles in the network to match the experimentally measured capillary pressure (Figure 12). We distribute contact angles at random to pores and throats in the model. We assume a uniform distribution of contact angle with a range of  $20^\circ$ . We then adjust the mean contact angle to obtain the best match to the capillary pressure; this is a one parameter fit. The waterflood capillary pressure is matched by assuming contact angles representative of a weakly oil-wet system, ranging between  $96^\circ$  and  $116^\circ$ ; see Figure 13. Note that the capillary pressure is entirely negative, showing that there is no spontaneous imbibition of water. This means that the injected water is behaving like a nonwetting phase.

[36] The advancing contact angles used in this work (that is contact angles for water displacing oil) are in line with the value of  $110^\circ$  inferred for this core plug by Masalmeh and Jing [2006]. They developed a mathematical method to calculate contact angles by scaling the primary drainage capillary pressure to derive imbibition capillary pressure.

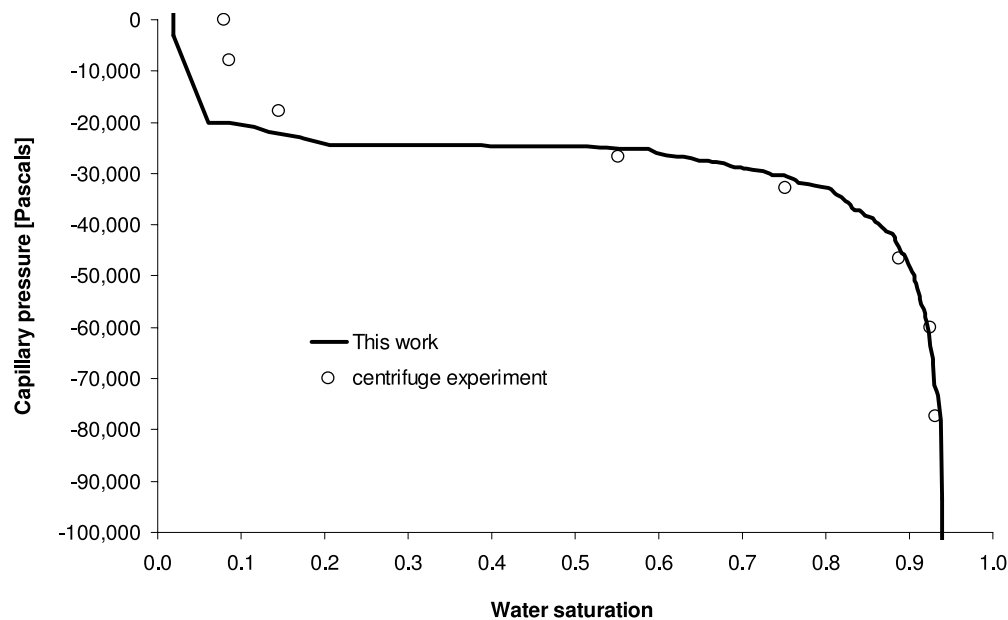


**Figure 11.** Comparison of network-predicted versus measured relative permeability during primary drainage.

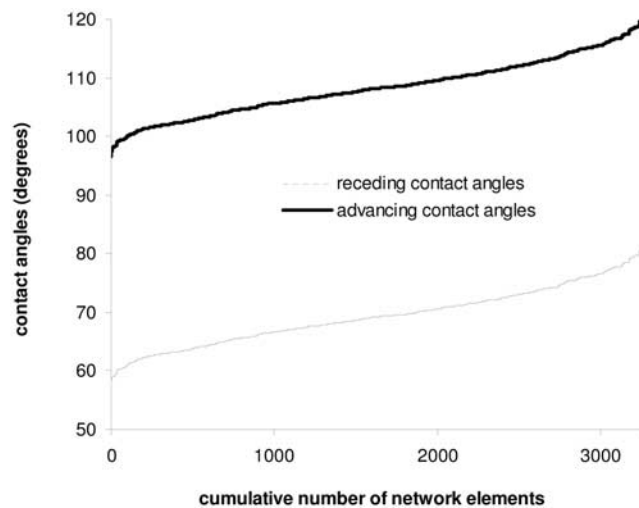
[37] The computed capillary pressure is a good match to the experiments, except at low water saturation. This could be due to the slow displacement of water at low imposed capillary pressure in the experiment, meaning that truly capillary equilibrium conditions have not been reached. Our prediction of the residual oil saturation, around 6%, is excellent. This value is lower than seen in water-wet sandstones, around 30% [Øren and Bakke, 2003; Valvatne and Blunt, 2004], for two reasons. First, because of the oil-wet nature of the rock, oil can form layers in the pore space

during waterflooding sandwiched between water in the corners and in the center [Valvatne and Blunt, 2004]. These layers provide connectivity of the oil down to very low saturation. Second, the high coordination number of the network allows phases to remain connected.

[38] After matching the waterflood capillary pressure we then make a prediction of the water and oil relative permeability as shown in Figure 14. The predicted oil relative permeability is in good agreement with the measured values.



**Figure 12.** Comparison of network-predicted versus measured waterflood capillary pressure (brine-oil centrifuge experiment). The contact angles in the network are adjusted to fit the measurements. A one-parameter fit is used where the contact angles are assigned at random with a range of  $20^\circ$  and the mean is varied. The best match, shown here, is with contact angles between  $96^\circ$  and  $116^\circ$ .



**Figure 13.** The network advancing and receding contact angles plotted against cumulative number of network elements.

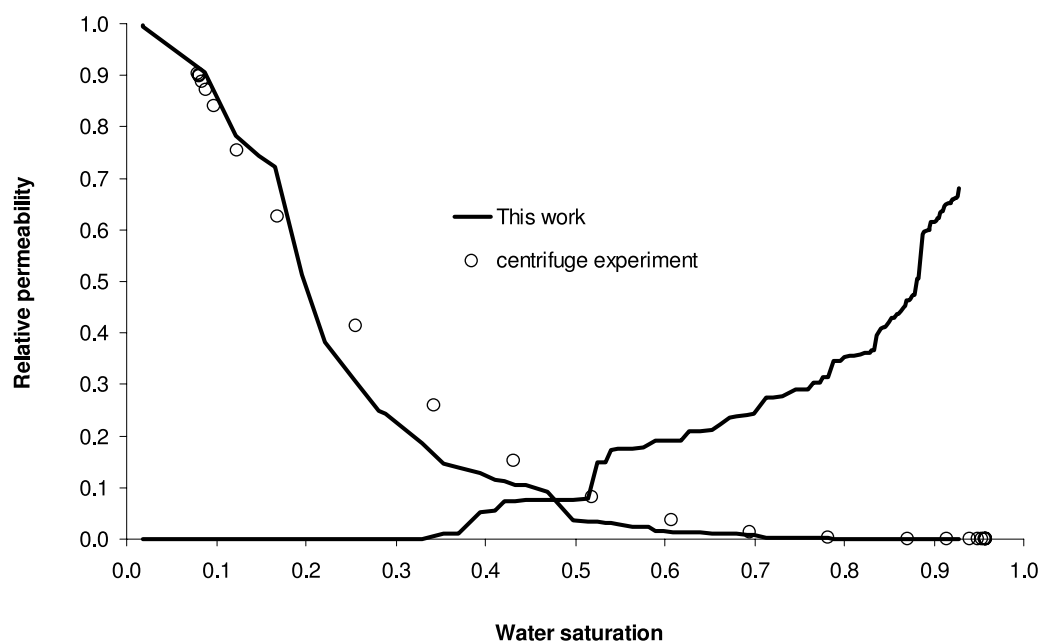
[39] Following waterflooding, we inject oil again. To match the experimental secondary drainage capillary pressure, we allow contact angle hysteresis (Figure 15). The hysteresis is given by a constant linear shift of  $37^\circ$  between the receding (oil displacing water) and advancing contact angles, shown in Figure 13. The receding contact angle estimated by *Masalmeh and Jing* [2006] for this core plug is around  $80^\circ$  which lies at the upper range of our contact angles. Note the significant degree of contact angle hysteresis: while the medium appears to be nonwetting to water during water injection (advancing contact angles greater than  $90^\circ$ ), it is also nonwetting to oil during oil injection (receding contact angles less than  $90^\circ$ ). This degree of

hysteresis is typical for neutrally wet or oil-wet systems [Dullien, 1992].

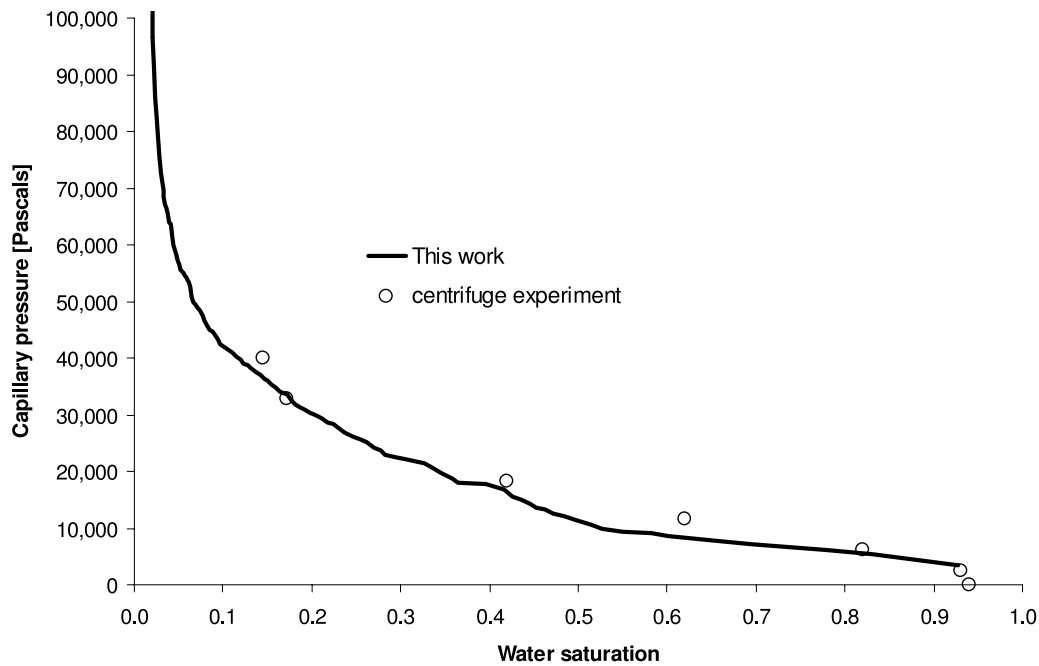
[40] The predicted relative permeability is given in Figure 16. The predicted water relative permeability is again in excellent agreement with the measurements.

[41] Figure 17 shows all the predicted relative permeabilities to highlight the computed hysteresis trends. During primary drainage, there is no trapping of water and, thanks to the high coordination number, both phases are well connected and the relative permeability is high. We would expect the water relative permeability to be higher for waterflooding, where water is the nonwetting phase, than in secondary oil invasion, where it is wetting; this is apparent in Figure 17 for water saturations less than around 0.8. Note that the oil relative permeability is very low beyond a water saturation of 0.5, even though oil continues to flow down to saturations of less than 0.1. The reason for this is that at low oil saturation, the oil is only connected in layers and occupies the centers of very few elements. While these layers do provide continuity of the oil, they have a very low conductance.

[42] The surprising finding is that the oil relative permeability in waterflooding (when oil is the wetting phase) is higher than in secondary oil invasion (when it is non-wetting). During waterflooding, layers of oil are formed sandwiched between water in the center of the pore space and water in the corners; it is these layers that maintain the connectivity of the oil phase down to low oil saturation. During secondary oil invasion, oil now displaces water from the larger pores. This can be done by piston-like displacement, or through the swelling of oil layers. Pores and throats can spontaneously fill with oil from layers as the oil pressure increases. While this leads to an increase in oil saturation, these elements are not well connected; they are only connected to other elements through low conductance layers, and so the relative permeability at a given oil saturation is lower than for both primary drainage (when



**Figure 14.** Comparison of network-predicted versus measured relative permeability during waterflood.



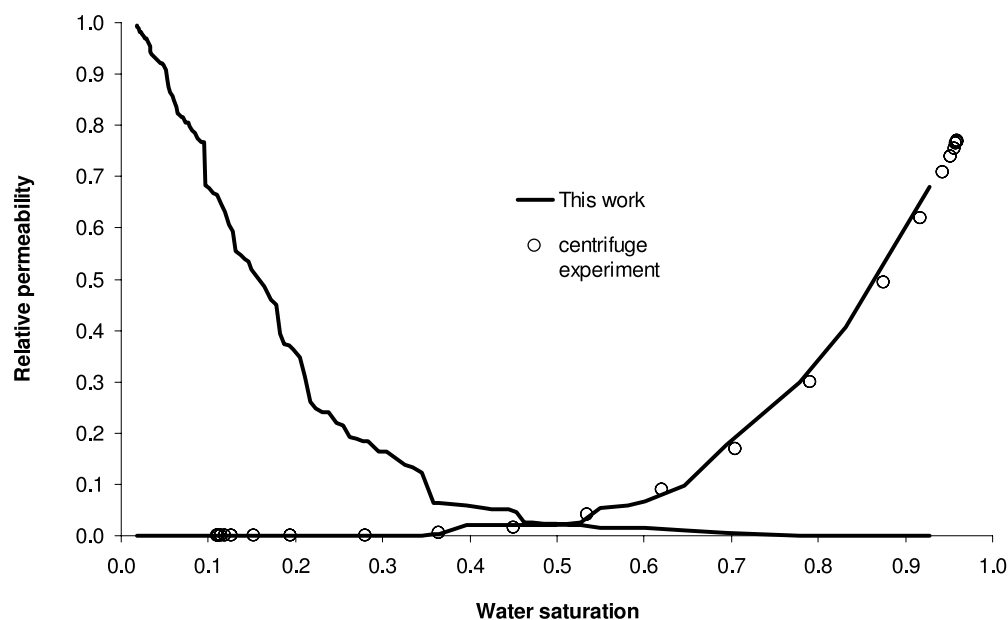
**Figure 15.** Comparison of network-predicted versus measured secondary oil injection capillary pressure (brine-oil centrifuge experiment).

the oil is always connected through the centers of elements) and waterflooding (when the oil is also connected through layers but the saturation for a given conductance is lower). There is no direct experimental evidence for this trend, however, since the centrifuge experiments only measure the relative permeability of the displaced phase.

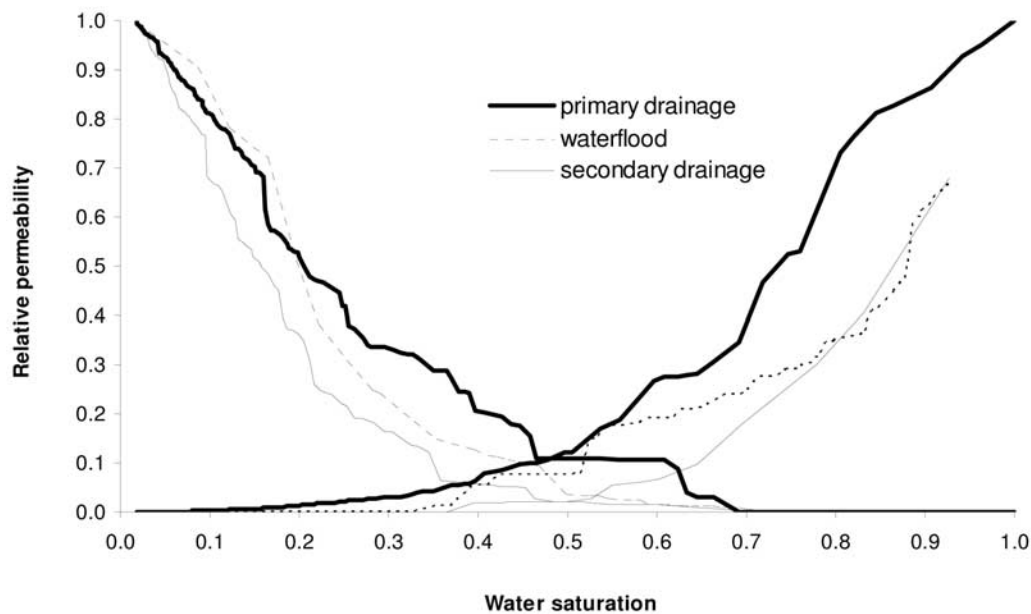
## 2.5. Sensitivity Analysis

[43] All the predictions shown so far have been based on a single network representing a tiny portion of the rock.

Furthermore, we have arbitrarily assigned shapes (or shape factors) to the network. To test the variability in our predictions from extracting networks from images of different plugs, Figure 18 shows the predicted secondary oil invasion relative permeabilities for three networks: plug A shows results from our original network, Figure 16; plug B is a network extracted from a different plug of similar permeability; and plug C is a third network from a different plug of similar permeability that contains vugs. There is significant variation in the predictions: similar variations are



**Figure 16.** Comparison of network-predicted versus measured relative permeability during secondary oil invasion.

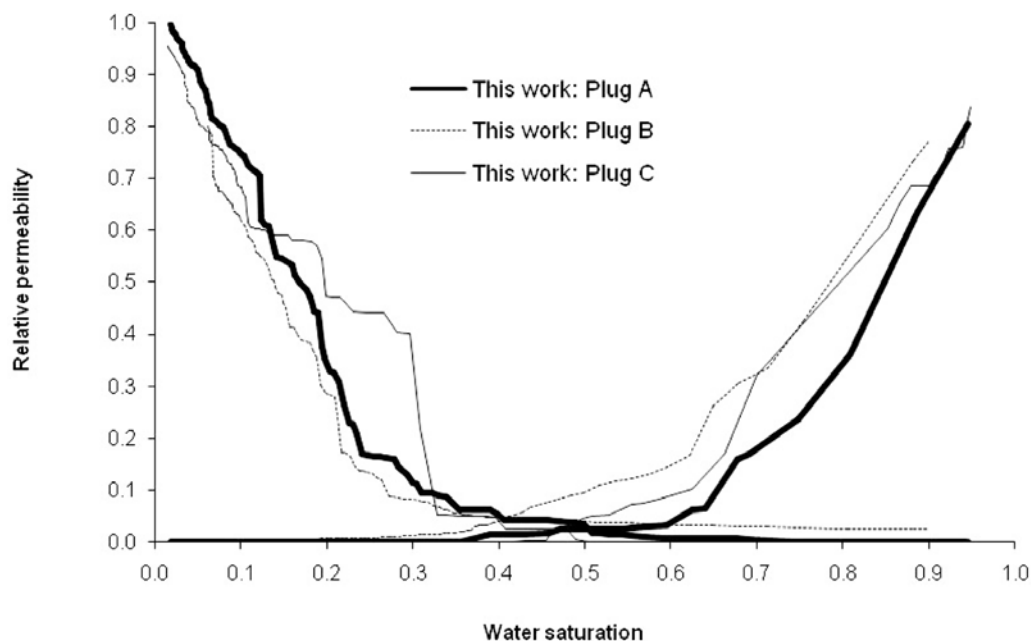


**Figure 17.** Comparison of network-predicted relative permeability data from all three cycles: primary drainage, waterflood, and secondary drainage.

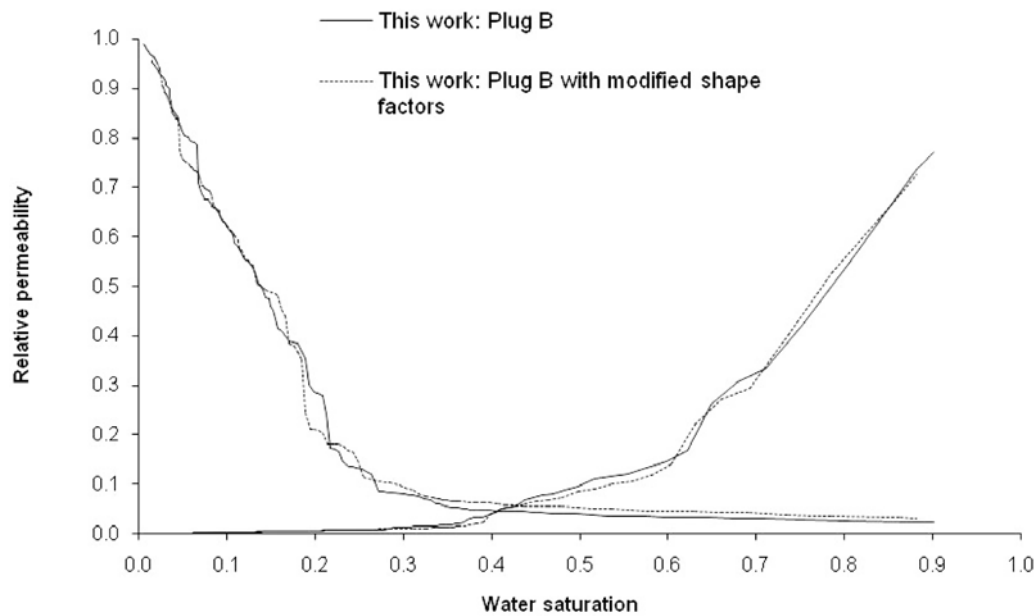
found with other networks and for the other relative permeabilities. In particular the vuggy network leads to relative permeability curves with sharp jumps when large pores become filled. The networks representing plugs B and C now overpredict the water relative permeability for this sample.

[44] To assess the sensitivity of the results to shape factor, Figure 19 compares the predicted secondary oil invasion

relative permeabilities for plug B with a distribution of shape factors and for a fixed value, 0.024, that is half that of an equilateral triangle. 0.024 is approximately the average value of the shape factor distribution applied in all the previous predictions. The results are similar for the two cases, indicating that the exact assignment of shape factor has little impact on the results: the results were insensitive to shape factor for the other networks and for other relative



**Figure 18.** Comparison of predicted relative permeabilities during secondary oil invasion. Plug A refers to the network whose results have been presented previously, plug B uses a network derived from another plug of similar permeability, and plug C has a similar permeability but the pore space contains vugs.



**Figure 19.** Comparison of predicted relative permeabilities for secondary oil invasion for plug B (see Figure 18). The same network topology is used in both cases, but in one a distribution of shape factors for each element is used (as in the previous results), while in the other the shape factor is kept fixed at half the value of an equilateral triangle (0.024).

permeability predictions, as long as we allowed fluid to be retained in the corners of most of the network elements. The conclusion of this section is that there is substantial variability in predictions from different networks, although the results are still consistent with the experimental measurements.

### 3. Discussion, Conclusions, and Recommendations for Future Work

[45] A network to represent the pore space was extracted from a subsurface carbonate core plug. A multistaged workflow was implemented where we started from a 2-D image and finished by predicting relative permeability following the assignment of contact angles to match the measured capillary pressures. We inferred a significant degree of contact angle hysteresis, such that the medium appeared nonwetting to water during water injection and nonwetting to oil during oil injection.

[46] The prediction of the primary drainage capillary pressure was excellent, indicating that our network adequately represented the pore space. With suitable contact angles, we predicted the waterflood residual oil saturation accurately as well as the relative permeability of oil for waterflooding and the relative permeability of water for secondary oil injection. We discovered a surprising hysteresis pattern in relative permeability that was explained by considering the pore-scale configuration of fluids.

[47] While these results are promising, the pore network model may not be representative of the rock investigated when large vugs or other geological variations are present. We saw large variations in predicted relative permeabilities when comparing networks extracted from different core samples with similar permeability. We were limited to constructing networks from images that contained only

$128^3$  voxels, representing a tiny sample of rock that might not be sufficiently large to include key geological features observed in the core plug sample investigated. Vugs and fractures can significantly affect fluid flow at the plug scale and network modeling will fail if these vugs are not properly incorporated into the model. For the systems studied in this work, vugs, although present, were not connected and no consistent impact on relative permeability was apparent.

[48] Further work could focus on generating several networks on the basis of different portions of a core plug, or different plugs, to assess the range of behavior that we might expect. Furthermore, larger networks should be generated, where computer resources permit, allowing a more representative model of the pore space to be generated.

[49] We do not propose a substitute to laboratory measurement; we recommend that pore network modeling should be used to understand the measurements and their associated uncertainties. Pore network modeling could be used to predict transport properties outside the range of parameters studied experimentally. In particular, this workflow could be useful to predict the relative permeability for a reservoir with variable wettability and pore structure across the field where experimental measurements are lacking.

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